II B. E (Electronics & Telecommunication)
Class Test III November 2023
Sub: 3ETRC2 Digital Electronics

Time: 70 min **Max Marks: 20**

Note: Attempt any four questions. All questions carry equal marks.

Q.1 Explain the working of negative edge triggered J-K flip-flop with the help of diagram containing negative edge detector circuit. 5

Q. 2 Design a digital circuit to convert RS flip-flop to T-Flip-flop. 5

Q. 3 Design a positive edge triggered mod-5 asynchronous counter using T-flip-flop. Also draw the timing diagram at output of all flip-flops. 5

Q. 4 A 5 bit switched current DAC has an output of 10mA for a digital input of 10100. What will be the output current for an digital input of 11010? 5

Q.5 Deign an A/D converter of 3 bit with maximum analog voltage of 7.0 V. The maximum quantization error allowed is ±0.5V. Also determine the input voltage at negative terminal of the comparators. Make suitable assumptions. 5